# A Distributional Minimax Law for Adaptive Threshold Estimation

Distribution-blind optimality, a sharp converse, and nested simulation

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**Abstract**

We estimate the population fraction $\theta = \mathbb{P}(P > 1/2)$ from adaptive Bernoulli observations of coins with latent success probabilities $P$. For an arbitrary magnitude law $\mu$ of $D = |P - 1/2|$ on $(0, 1/2]$, define $\omega_\mu(h) = \mathbb{E}_\mu \min\{1, h/D\}$. The minimax root-mean-square error under a uniform expected observation budget $B$ is $\asymp \omega_\mu(B^{-1/2})$, with universal constants; the high-confidence radius is $\asymp \omega_\mu(\sqrt{\log(1/\delta)/B})$. A single algorithm knowing only $B$ (and $\delta$ for high confidence) attains these rates simultaneously for every $\mu$, without observing individual magnitudes. The lower bounds allow arbitrary revisiting and interleaving, even with revealed or selected magnitudes. The same results hold over stochastic-dominance envelope classes. An unbiased estimator can have both uniformly finite variance $V$ and expected work $W$ over the sign class exactly when $I_\mu = \mathbb{E}_\mu[D^{-1}] < \infty$. Every such estimator satisfies $VW \geq I_\mu^2/16$, and the universal coefficient $1/16$ is sharp. A parameter-free randomized multilevel estimator achieves $VW \leq 12665 I_\mu^2$. A continuous-magnitude observed-state construction satisfies the bounded-density and conditional-moment assumptions of adaptive nested simulation and yields the matching $\varepsilon^{-2}\log^2(1/\varepsilon)$ lower rate. Polynomial, logarithmic, and iterated-log margin laws follow as corollaries; the critical obstruction also persists under fixed finite-order density smoothness. Distribution-blind risk control does not provide an observable confidence radius or a deterministic work cap.

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## 1 Introduction and contribution

Estimating the fraction of a population lying above a latent threshold is not the same task as classifying every member of that population. Observations can be concentrated on promising or ambiguous members, and biased individual classifications can be combined through telescoping estimators. These possibilities make algorithm-independent lower bounds important.

We study the following basic observation model. A fresh coin has a latent probability $P$ drawn from a population distribution $G$. Conditional on $P$, its observations are independent Bernoulli trials. An algorithm may adapt to its entire history: it can create new coins, interleave observations of old coins, and stop at a random time. Its target is

$$\theta(G) = \mathbb{P}_G(P > 1/2). \tag{1}$$

One coin flip is one unit of work. The principal obstruction is the population's mass near $1/2$, but a single power exponent need not describe that mass accurately.

**Distributional and distribution-blind optimality.** The central object is the entire function

$$\omega_\mu(h) = \int \min\{1, h/d\}\, \mu(\mathrm{d}d), \qquad \mu = \mathcal{L}(|P - 1/2|). \tag{2}$$

The main theorem identifies this function as the minimax error law. Thus irregular, discrete, and logarithmically modified margin distributions are covered by one statement, rather than by separate exponent calculations. The same algorithm attains the law for every magnitude distribution, rather than requiring a distribution-dependent allocation. A second theorem characterizes unbiased feasibility by an inverse moment and proves a sharp universal lower constant for the variance–work product. Writing $F_\mu(h) = \mathbb{P}(D \le h)$ makes the two parts of the modulus explicit:

$$\omega_\mu(h) = F_\mu(h) + h\mathbb{E}_\mu[D^{-1}; D > h].$$

The first term represents unresolved near-threshold mass; the second records the accumulated contribution of the remaining difficulty scales.

The lower bound makes many difficulty scales jointly hard: each scale contributes to the target separation, while every next observation has uniformly bounded information. The upper bound is a fixed geometric randomization of capped sequential classifiers. Its conditional second moment and expected work are both controlled by the same truncated inverse margin. This common conditional bound is what permits arbitrary margin laws. An exact two-success inverse-binomial calibration then chooses the number of production samples without estimating either their mean cost or variance. The underlying inverse-sampling identity is classical [8]; its role here is to eliminate all margin knowledge from the budget-indexed rule.

## 1.1 Relationship to established work

**Nested simulation.** Uniform inner-sampling analyses include Lee–Glynn [10] and Gordy–Juneja [11]; Broadie, Du, and Moallemi [9] also identify Lee's 1998 thesis as an early analysis and introduce sequential inner allocation for loss-probability estimation. Giles and Haji-Ali [2] combine adaptive inner sampling and multilevel Monte Carlo (MLMC) to attain $O(\varepsilon^{-2}|\log\varepsilon|^2)$ work for thresholded nested expectations. Haji-Ali, Spence, and Teckentrup [12] develop a broader adaptive MLMC framework for probabilities. We do not claim these methods or the critical upper rate as new. Our observed-state construction in theorem 18 verifies Assumptions 2.1 and 2.4 of [2] literally and proves the matching lower rate on an unstructured subclass. It does not address models with additional smooth cross-state structure or alternative information oracles. Giles's survey [13] discusses the high kurtosis of level corrections for discontinuous functions; our confidence amplification uses only second moments, not an empirical variance estimate or a normal approximation.

**Adaptive coin mixtures and information bounds.** Lee and Valiant [1] prove fully adaptive optimal bounds for mixtures separated from 1/2 by a fixed gap, including dependence on the positive fraction. Our gap-class specialization agrees with their rate in the balanced regime (corollary 14). The lower bounds here are driven by sign priors near 1/2; this explains why a uniform local posterior inequality and the stopped KL chain rule suffice. Their more refined rare-event analysis is not reproduced by our supremum over all sign functions. The change-of-measure step is related to the standard sequential information inequality of Kaufmann, Cappé, and Garivier [14]; it is not a new general principle. Lee–Valiant also optimize estimators for known positive and negative conditional bias laws under a per-coin depth constraint. Here the magnitude law is arbitrary, the sign function is unrestricted, and our lower bounds have no per-coin depth constraint.

**Randomization and unbiased estimation.** Single-term multilevel debiasing is established methodology [3, 4]. Blanchet and Glynn [15] obtain finite work-normalized variance under regularity such as local twice differentiability of the outer function. Syed and Wang [5] study repeatedly nested expectations under suitable regularity conditions. Our threshold indicator is discontinuous: the inverse-margin condition and all-algorithm converse characterize what remains possible in this setting. The inverse-binomial sampling identity used for cost calibration dates to classical work of Haldane [8]; we give a self-contained proof and do not claim that identity as new.

**Other observation models and scope.** Brennan, Vinayak, and Jamieson [6] estimate mass above effect-size thresholds from noisy test statistics, rather than unrestricted sequential revisiting. Chen, Naslidnyk, and Briol [7] study nested expectations with kernel quadrature and identify minimax lower bounds as a research direction. Their smooth-kernel setting differs from ours. The proposed contribution is the arbitrary-margin minimax characterization, a distribution-blind optimal-rate algorithm, and the inverse-margin unbiased converse in the precisely specified experiment. These statements are not claims of instance optimality, rare-event optimality, or a general solution to all nested-expectation problems.

## 2 Model, notation, and main results

Fix $a = 1/2$. Let $\mu$ be any probability measure on $(0, a]$; in particular, $\mu$ may have atoms, gaps, or a singular continuous part. A measurable function $q : (0, a] \to [0, 1]$ specifies a population by

$$D \sim \mu, \qquad S \mid D = d \sim \mathrm{Ber}(q(d)), \qquad P = \tfrac{1}{2} + (2S - 1)D. \tag{3}$$

The variables attached to distinct new coins are independent. The population class is $\mathcal{G}(\mu) = \{G_{\mu,q} : q \text{ measurable}\}$ and $\theta(\mu, q) = \int q(d)\,\mu(\mathrm{d}d)$. Individual magnitudes and signs are not observed. The upper algorithm does not know $\mu$ either. The distribution-wise minimax benchmark may grant knowledge of $\mu$, so attaining it simultaneously without that knowledge is the stronger statement. The lower bound remains valid with still more information. The endpoint $D = 1/2$ is allowed and corresponds to deterministic coins; $D = 0$ is excluded.

An algorithm has an independent random seed, creates finitely many coins before each observation, selects one of its coins, and receives its next Bernoulli observation. It stops after $T$ observations and returns a real number $\widehat{\theta}$. Infinite expected work is permitted in the model, but algorithms satisfying a finite uniform budget necessarily stop almost surely. Finite internal computation between observations is implicit. We count observations, not arithmetic operations or independent random bits. Every coin actually created by the upper algorithm receives at least one observation. Charging a fixed additional amount per new outer draw therefore changes its cost by at most a constant factor.

Define

$$R_\mu(B) = \inf_{\mathcal{A}:\, \sup_q \mathbb{E}_{\mu,q} T \le B} \sup_q \big(\mathbb{E}_{\mu,q}(\widehat{\theta} - \theta)^2\big)^{1/2}, \tag{4}$$

$$r_\mu(B, \delta) = \inf_{\mathcal{A}:\, \sup_q \mathbb{E}_{\mu,q} T \le B} \inf\{r \ge 0 : \sup_q \mathbb{P}_{\mu,q}(|\widehat{\theta} - \theta| > r) \le \delta\}. \tag{5}$$

These are expected-budget criteria, not deterministic maximum-work criteria.

For $h > 0$, write

$$A_\mu(h) = \mathbb{E}_\mu \min\{h^{-1}, D^{-1}\}, \qquad \omega_\mu(h) = hA_\mu(h), \qquad I_\mu = \mathbb{E}_\mu[D^{-1}] \in [2, \infty]. \tag{6}$$

The modulus is increasing and concave, $\omega_\mu(0+) = 0$, and $\omega_\mu(h) = 1$ for $h \geq a$, while $\omega_\mu(h) \geq \min\{1, 2h\}$. In particular, for $c \geq 1$ and $0 < b \leq 1$,

$$\omega_\mu(ch) \leq c\omega_\mu(h), \qquad \omega_\mu(bh) \geq b\omega_\mu(h). \tag{7}$$

**Theorem 1** (Distribution-blind arbitrary-margin minimax law)**.** *There are universal constants $0 < c < C < \infty$ such that for every probability measure $\mu$ on $(0, 1/2]$, every $B \geq 1$, and every $0 < \delta \leq 1/4$,*

$$c\omega_\mu(B^{-1/2}) \leq R_\mu(B) \leq C\omega_\mu(B^{-1/2}), \tag{8}$$

$$c\omega_\mu\left(\sqrt{\frac{\log(1/\delta)}{B}}\right) \leq r_\mu(B, \delta) \leq C\omega_\mu\left(\sqrt{\frac{\log(1/\delta)}{B}}\right). \tag{9}$$

*More strongly, for each $B$ there is one algorithm $\mathcal{A}_B$, depending only on $B$, that simultaneously for all $\mu, q$ satisfies*

$$\mathbb{E}_{\mu,q} T \leq B, \qquad \left(\mathbb{E}_{\mu,q}(\widehat{\theta} - \theta)^2\right)^{1/2} \leq C\omega_\mu(B^{-1/2}). \tag{10}$$

*There is likewise one algorithm depending only on $(B, \delta)$ for the upper confidence bound. Neither algorithm observes $D$ or knows $\mu$. In the lower bounds one may take $c = 1/64$ for RMS and $c = 15/(256\sqrt{2})$ for confidence. The lower bounds hold even if every coin's magnitude is revealed for free, or if the algorithm selects the magnitude of each fresh coin. In the latter experiment $q$ is defined on all $(0, 1/2]$ and the target is still $\int q \, \mathrm{d}\mu$.*

This is distribution-wise minimax optimality: the worst case is over all sign functions at a fixed magnitude law. It is not instance optimality for each particular pair $(\mu, q)$. The budget is expected work, not an almost-sure cap. The modulus in the guarantee is evaluated at the true, possibly unknown distribution; it is not an observable confidence radius.

**Theorem 2** (Exact unbiased feasibility and variance–work scale)**.** *An estimator unbiased for every $G \in \mathcal{G}(\mu)$ can have both*

$$V = \sup_q \mathrm{Var}_{\mu,q}(Y) < \infty, \qquad W = \sup_q \mathbb{E}_{\mu,q} T < \infty$$

*if and only if $I_\mu < \infty$. When $I_\mu < \infty$,*

$$\frac{1}{16} I_\mu^2 \;\leq\; \inf_{Y \text{ unbiased}} VW \;\leq\; 12665 I_\mu^2, \tag{11}$$

*where the infimum is over estimators with both uniform bounds finite. A single estimator, using neither $\mu$ nor any margin parameters, attains the upper bound simultaneously for every such magnitude law. It satisfies $\sup_q \mathbb{E}Y^2 \leq 85 I_\mu$ and $\sup_q \mathbb{E}T \leq 149 I_\mu$. The lower coefficient $1/16$ is the exact universal constant: equality is attained for every fixed-gap law $\mu = \delta_d$. Equality is not asserted for arbitrary nondegenerate $\mu$.*

## 2.1 Unknown magnitude laws under an envelope

Let $F_\mu(t) = \mu((0, t])$, extended to equal one above $a$, and define

$$\mathcal{G}_{\preceq \mu} = \{G \text{ on } [0,1] : \mathbb{P}_G(|P - 1/2| \leq t) \leq F_\mu(t) \text{ for every } t > 0\}. \tag{12}$$

Here the actual magnitude law is unknown on $(0, 1/2]$. The inequality means its magnitudes are stochastically no smaller than those under $\mu$. It also rules out an atom at $P = 1/2$.

**Corollary 3** (Stochastic-dominance envelope version)**.** *Both conclusions of theorem 1 hold with the supremum taken over $\mathcal{G}_{\preceq \mu}$, with the same universal constants. The existence criterion and variance–work product bounds of theorem 2 also hold for this class. The finite-budget and unbiased algorithms need not know the envelope; it is used only to express a uniform guarantee.*

**What distribution-blindness does not supply.** The algorithm is specified without $\mu$ or an envelope. A numerical choice of budget for a prescribed accuracy, however, generally requires a bound on the unknown modulus. Neither the adaptive risk guarantee nor the unbiased estimator automatically supplies a data-computable confidence radius.

**Corollary 4** (Critical nested-simulation obstruction)**.** *Let $U \sim \mathrm{Unif}[0,1]$, $X \in \{-1,1\}$, and $D(U) = |\mathbb{E}[X \mid U]|/2$. Consider all measurable conditional models with $D(U) \sim \mathrm{Unif}(0,1/4]$, with no cross-state regularity. For estimating $\mathbb{P}(\mathbb{E}[X \mid U] > 0)$, the minimax expected observation cost at RMS error $\varepsilon$, or at error $\varepsilon$ with a fixed failure probability $\delta_0 = 1/16$, is*

$$\asymp \varepsilon^{-2} \log^2(1/\varepsilon), \qquad \varepsilon \downarrow 0.$$

*The lower bound allows arbitrary selected-state queries and revisiting. Every model satisfies Assumptions 2.1 and 2.4 of Giles–Haji-Ali [2], with uniform constants.*

The continuous-magnitude construction and proof are given in theorem 18. This is a worst-case sample-oracle statement on a subclass satisfying those assumptions, not a claim about every structured simulator or every possible source of information.

## 3 A history-uniform adaptive information bound

For Bernoulli probabilities $u, v \in (0,1)$,

$$\mathrm{kl}(u,v) \le \frac{(u-v)^2}{v(1-v)}. \tag{13}$$

This follows from $\log x \le x - 1$ applied to the two likelihood ratios. We next control how an arbitrary local observation history changes two symmetric sign priors.

**Lemma 5** (Posterior contraction at every magnitude)**.** *Fix $0 < d \le \frac{1}{2}$ and $0 \le \eta < \frac{1}{2}$. Under two experiments the sign priors are $q_\pm = \frac{1}{2} \pm \eta$, while the observation likelihoods conditional on the sign are the same. Let $L \in [0,\infty]$ be the likelihood ratio of an arbitrary local history of one coin under positive versus negative sign, let $\rho_\pm$ be its posterior sign probabilities, and let $p_\pm = \frac{1}{2} - d + 2d\rho_\pm$ be the next predictive success probabilities. If $0 < L < \infty$, then*

$$\rho_+ - \rho_- = \frac{2\eta L}{L + (\frac{1}{4} - \eta^2)(L-1)^2} \le 2\eta. \tag{14}$$

*In every case,*

$$\mathrm{kl}(p_+, p_-) \le \frac{32\, d^2 \eta^2}{(1-4\eta^2)^2}. \tag{15}$$

*The same bound holds for $\mathrm{kl}(p_-, p_+)$. The constant 32 is sharp: at $L = 1$ the ratio of the two sides of (15) tends to one as $\eta \downarrow 0$.*

*Proof.* If $L \in \{0,\infty\}$, then $\rho_+ = \rho_- \in \{0,1\}$, so $p_+ = p_-$ and the divergence vanishes. This case cannot occur for $d < \frac{1}{2}$; for $d = \frac{1}{2}$ it occurs after the first observation. Now let $0 < L < \infty$ and write $a = \frac{1}{2} + \eta$, $b = \frac{1}{2} - \eta$. Then $a + b = 1$ and $ab = \frac{1}{4} - \eta^2 = (1 - 4\eta^2)/4$. Bayes' formula gives $\rho_+ = aL/(b + aL)$ and $\rho_- = bL/(a + bL)$, so

$$\rho_+ - \rho_- = \frac{(a^2 - b^2)L}{(b + aL)(a + bL)}, \qquad (b + aL)(a + bL) = ab(L-1)^2 + (a+b)^2 L.$$

This proves (14). Also

$$\rho_-(1-\rho_-) = \frac{abL}{(a+bL)^2}, \qquad \rho_+(1-\rho_+) = \frac{abL}{(b+aL)^2}.$$

For fixed $v \in (0,1)$, the function $u \mapsto \mathrm{kl}(u,v)$ vanishes together with its first derivative at $u = v$, and its second derivative is $1/\{u(1-u)\}$. Since $p_- \le p_+$, Taylor's theorem gives

$$\mathrm{kl}(p_+, p_-) \le \frac{(p_+ - p_-)^2}{2\min_{\xi\in[p_-,p_+]} \xi(1-\xi)}.$$

The same Taylor upper bound applies with the two arguments interchanged, since its right side depends only on their squared difference and the interval between them. Thus it controls both KL directions. Every such $\xi$ equals $\frac{1}{2} - d + 2d\rho$ for some $\rho \in [\rho_-, \rho_+] \subset (0,1)$. Because $d \le \frac{1}{2}$,

$$\xi(1-\xi) = \tfrac{1}{4} - d^2(2\rho-1)^2 \ge \tfrac{1}{4}\{1-(2\rho-1)^2\} = \rho(1-\rho).$$

By concavity, the minimum of $\rho(1-\rho)$ over $[\rho_-, \rho_+]$ is attained at an endpoint. Since $p_+ - p_- = 2d(\rho_+ - \rho_-)$, the preceding displays give

$$\mathrm{kl}(p_+, p_-) \le 2d^2 \max\Big\{\frac{(\rho_+-\rho_-)^2}{\rho_-(1-\rho_-)}, \frac{(\rho_+-\rho_-)^2}{\rho_+(1-\rho_+)}\Big\} = \frac{8d^2\eta^2}{ab}\max\Big\{\frac{L}{(b+aL)^2}, \frac{L}{(a+bL)^2}\Big\}.$$

The maps $L \mapsto L/(b+aL)^2$ and $L \mapsto L/(a+bL)^2$ are maximized at $L = b/a$ and $L = a/b$ respectively, both with value $1/(4ab)$. Therefore $\mathrm{kl}(p_+, p_-) \le 2d^2\eta^2/(ab)^2$, which is (15).

For sharpness, at $L = 1$ we have $p_\pm = \frac{1}{2} \pm 2d\eta$ and

$$\mathrm{kl}(p_+, p_-) = 4d\eta \log\frac{1+4d\eta}{1-4d\eta} = 32d^2\eta^2\{1 + O(d^2\eta^2)\}. \qquad \square$$

Fix $h > 0$ and take

$$\eta_h(d) = \frac{1}{8}\min\{1, h/d\}, \qquad q_\pm(d) = \frac{1}{2} \pm \eta_h(d). \tag{16}$$

The two targets are centered at $1/2$ and separated by

$$\Delta_h = \theta_+ - \theta_- = 2\mathbb{E}_\mu \eta_h(D) = \frac{1}{4}\omega_\mu(h). \tag{17}$$

Because $d\eta_h(d) \le h/8$, every next observation has predictive KL at most $128h^2/225$ in either direction.

**Proposition 6** (Stopped-transcript bound)**.** *For every adaptive algorithm in any of the experiments of theorem 1, its stopped transcript laws satisfy*

$$\mathrm{KL}(\mathsf{P}_+^{\mathcal{A}}, \mathsf{P}_-^{\mathcal{A}}) \le \frac{128}{225} h^2 \mathbb{E}_+ T. \tag{18}$$

*Consequently, if it estimates both targets to radius $\varepsilon < \Delta_h/2$ with failure probability at most $\delta < 1/2$,*

$$\mathbb{E}_+ T \ge \frac{225}{128h^2}\,\mathrm{kl}(1-\delta, \delta). \tag{19}$$

*Proof.* First reveal all magnitudes; an algorithm not using them is a special case. Conditional on a realized transcript, action-selection and randomization factors are common to the two hypotheses. Conditional on the hypothesis and revealed magnitudes, the posterior of a coin's

sign depends only on its own observations: the likelihood factorizes by coin. Thus the local posterior identity applies even after arbitrary revisiting and interleaving. For a fresh coin the same assertion has $L = 1$. At $d = 1/2$ the first observation has full support under both hypotheses, and after it the sign is revealed, so later observations contribute zero KL.

A newly drawn magnitude has the same law in both experiments and contributes zero KL. A selected magnitude is an algorithmic action and also contributes zero conditional KL. The chain rule, first stopped at $T \wedge m$, bounds transcript KL by $128h^2\mathbb{E}_+(T \wedge m)/225$. Padding the transcript after its stopping time and taking increasing finite prefixes gives (18) by monotone convergence of expected work and the finite-prefix characterization of relative entropy. If expected work is infinite, the bound is vacuous. Interchanging the hypotheses gives the reverse bound with $\mathbb{E}_- T$.

Thresholding $\widehat{\theta}$ at $1/2$ gives a test with both errors at most $\delta$, since the estimation radius is strictly less than half the separation. Data processing gives $\mathrm{KL} \geq \mathrm{kl}(1-\delta, \delta)$, proving (19). □

The strict separation in this proposition is intentional. At exactly half the separation, the constant output $1/2$ is within tolerance for both members of the pair, so a test cannot be inferred without a limiting argument over strictly separated pairs.

## 4 A universal capped classifier

This section does not assume a magnitude law. Fix a coin with $P = 1/2 + (2S-1)d$, where $0 < d \leq 1/2$ and $S \in \{0, 1\}$. For a dyadic cap $N = 2^\ell$, examine prefixes of lengths $1, 2, 4, \ldots, N$. At a prefix $n < N$, stop when

$$\left|\overline{X}_n - \frac{1}{2}\right| > t_{n,N}, \qquad t_{n,N} = \sqrt{\frac{1 + \log(N/n)}{n}}, \tag{20}$$

and output the side of $1/2$ occupied by the sample mean. If no early stop occurs, output $Z_N = \mathbf{1}\{\overline{X}_N > 1/2\}$; a final tie is negative. Let $\tau_N$ be the number of observations used.

Set

$$m_N(d) = \min\{N, d^{-2}[1 + \log_+(Nd^2)]\}, \qquad \log_+ x = \max\{0, \log x\}. \tag{21}$$

**Lemma 7** (Conditional error and work)**.** *For every dyadic $N \geq 1$, both signs, and $0 < d \leq 1/2$,*

$$e_N(d, S) := \mathbb{P}(Z_N \neq S \mid d, S) \leq \min\{1, (Nd^2)^{-2}\}, \tag{22}$$

$$\mathbb{E}[\tau_N \mid d, S] \leq 17 m_N(d). \tag{23}$$

*Proof.* A wrong early stop at a fixed prefix $n$ requires a one-sided deviation of at least $d + t_{n,N}$. Hoeffding's inequality and a union bound over fixed prefixes, not a conditioning on the stopping event, give

$$\begin{aligned} e_N(d, S) &\leq e^{-2Nd^2} + \sum_{\substack{n<N \\ n \text{ dyadic}}} e^{-2n(d+t_{n,N})^2} \\ &\leq e^{-2Nd^2} + e^{-2} \sum_{\substack{n<N \\ n \text{ dyadic}}} (n/N)^2 e^{-2nd^2}. \end{aligned} \tag{24}$$

For every dyadic sequence of positive $u$ values and every $v, b > 0$,

$$\sum_u u^v e^{-bu} \leq \frac{\Gamma(v+1)}{(1-2^{-v})b^v}. \tag{25}$$

Indeed, the disjoint intervals $[u/2, u]$ satisfy $\int_{u/2}^{u} y^{v-1}e^{-by}\,\mathrm{d}y \geq e^{-bu}u^v(1-2^{-v})/v$. With $x = Nd^2 \geq 1$, the sum in (24) is bounded by $x^{-2}(2/3)$, and $x^2e^{-2x} \leq e^{-2}$. Hence the right side is at most $(5/3)e^{-2}x^{-2} < x^{-2}$. The bound by one handles $x < 1$.

The dyadic survival identity and the fact that a surviving prefix cannot have crossed its threshold imply

$$\begin{aligned}\mathbb{E}\tau_N &= 1 + \sum_{n<N,\, n \text{ dyadic}} n\mathbb{P}(\tau_N > n) \\ &\leq 1 + \sum_{n<N,\, n \text{ dyadic}} n\{\mathbf{1}[d \leq 2t_{n,N}] + 2e^{-nd^2/2}\}. \end{aligned} \tag{26}$$

For $x \geq 1$, the inequality $d \leq 2t_{n,N}$ implies $nd^2 \leq 4(1+\log x)$: if $nd^2 < 1$ this is immediate, and otherwise $\log(N/n) = \log(x/(nd^2)) \leq \log x$. The corresponding dyadic sum of $n$ is at most $8d^{-2}(1+\log x)$. Using (25) with $v = 1, b = 1/2$ bounds the exponential part of (26) by $8d^{-2}$. Since $1 \leq d^{-2}$, the total is at most $17d^{-2}(1+\log x)$. Also $\tau_N \leq N$ deterministically, and $1 + \log x \leq x$ for $x \geq 1$. For $x < 1$, $m_N(d) = N$, completing the proof. □

### 4.1 One geometric level law for every population

On a common coin stream define

$$\Delta_0 = Z_1, \qquad \Delta_\ell = Z_{2^\ell} - Z_{2^{\ell-1}} \quad (\ell \geq 1). \tag{27}$$

A coupled difference is computed by running both classifiers on the same prefix until both have stopped. Its work $W_\ell$ is the maximum of their stopping times and hence at most their sum. Let

$$r = 2^{-1/2}, \qquad p_\ell = (1-r)r^\ell, \qquad \mathbb{P}(J = \ell) = p_\ell. \tag{28}$$

Choose $J$ independently of a fresh coin and its observations. For a finite cutoff $L$, return

$$Y_L = \frac{\Delta_J}{p_J}\mathbf{1}\{J \leq L\}. \tag{29}$$

When $J > L$, no coin is sampled and the output and work are both zero. Write $\mathcal{W}_L$ for actual work. For $L = \infty$, drop the indicator and write $Y, \mathcal{W}$.

**Lemma 8** (Truncated inverse-margin control)**.** *For every finite $L \geq 0$, $N = 2^L$, both signs, and $0 < d \leq 1/2$,*

$$\mathbb{E}[Y_L \mid d, S] = \mathbb{E}[Z_N \mid d, S], \tag{30}$$

$$\mathbb{E}[Y_L^2 \mid d, S] \leq 85\min\{\sqrt{N}, d^{-1}\}, \tag{31}$$

$$\mathbb{E}[\mathcal{W}_L \mid d, S] \leq 149\min\{\sqrt{N}, d^{-1}\}. \tag{32}$$

*Moreover,*

$$\sum_{\ell \geq 0}\mathbb{E}[|\Delta_\ell| \mid d, S] \leq C_0[1 + \log(1/d)] \tag{33}$$

*for a universal $C_0$. Whenever $\mathbb{E}\log(1/D) < \infty$, $Y$ is integrable and unbiased for $\theta$. This logarithmic condition is sufficient here; no necessity claim for integrability alone is made. If $\mathbb{E}D^{-1} < \infty$, its second moment is at most $85\mathbb{E}D^{-1}$ and its expected work at most $149\mathbb{E}D^{-1}$.*

*Proof.* The finite expectation telescopes. For $\ell \geq 1$, a nonzero difference requires at least one classifier to be wrong, so, with $N_\ell = 2^\ell$,

$$\mathbb{E}[\Delta_\ell^2 \mid d, S] \leq 5\min\{1, (N_\ell d^2)^{-2}\}, \qquad \mathbb{E}[W_\ell \mid d, S] \leq 34 m_{N_\ell}(d). \tag{34}$$

The same loose inequalities cover $\ell = 0$. Consequently

$$\mathbb{E}[Y_L^2 \mid d, S] \le \frac{5}{1-r} \sum_{\ell=0}^{L} \sqrt{N_\ell} \min\{1, (N_\ell d^2)^{-2}\}, \tag{35}$$

$$\mathbb{E}[\mathcal{W}_L \mid d, S] \le 34(1-r) \sum_{\ell=0}^{L} N_\ell^{-1/2} m_{N_\ell}(d). \tag{36}$$

Split each sum at $N_\ell = d^{-2}$. Before that point, both summands are $\sqrt{N_\ell}$ and their sum is bounded by $(1-r)^{-1} \min\{\sqrt{N}, d^{-1}\}$. After that point, the second-moment sum is a geometric tail with ratio $2^{-3/2}$ and is at most $(1-r^3)^{-1} d^{-1}$. Its full constant is bounded by

$$\frac{5}{1-r}\left(\frac{1}{1-r} + \frac{1}{1-r^3}\right) < 85.$$

For work, the first $N_\ell$ after the split has $1 < N_\ell d^2 \le 2$. Its tail is bounded by

$$d^{-1} \sum_{k \ge 0} r^k (1 + \log 2 + k \log 2).$$

Combining the prefix and tail gives the constant

$$34\left(2 + \log 2 + \frac{r \log 2}{1-r}\right) < 149.$$

If the cutoff precedes the split, only the smaller prefix bound is needed. This proves (31)–(32).

Because $|\Delta_\ell| = \Delta_\ell^2$ for $\ell \ge 1$, the number of levels before the split is $O(1 + \log(1/d))$, and the subsequent error bounds form a convergent geometric series. This proves (33). Absolute summability allows Fubini and telescoping to $S$, using (22). Averaging over $(D, S)$ proves unbiasedness whenever the stated logarithmic moment is finite. The inverse-moment condition implies that condition. Finally take $L \to \infty$ in the nonnegative series for second moment and work. □

This is the standard single-term randomized multilevel identity [4]; the common conditional bounds and their distributional consequences are the key additional estimates. The estimator's output need not lie in $[0, 1]$. The interior-magnitude obstruction to nonnegative output is given in section 6.1.

# 5 Proof of the minimax law

## 5.1 Exact calibration without a known margin law

For $h = N^{-1/2}$ define

$$b_\mu(h) = \mathbb{E}_\mu \min\{1, (h/D)^4\} \le \omega_\mu(h). \tag{37}$$

One call of the cutoff estimator returns $(Y_L, \mathcal{W}_L)$ with

$$0 \le \mathcal{W}_L \le N, \quad c := \mathbb{E}\mathcal{W}_L \le 149 A_\mu(h), \quad v := \mathrm{Var}(Y_L) \le 85 A_\mu(h), \quad |\mathbb{E}Y_L - \theta| \le b_\mu(h). \tag{38}$$

Moreover $c > 0$, because level zero has probability $1 - r > 0$ and costs one observation. The following wrapper handles the unknown $c$ exactly.

**Lemma 9** (Two-success cost calibration)**.** *Suppose an independent call returns $(Z, C)$ with $0 \le C \le N$, $c = \mathbb{E}C > 0$, $\mathbb{E}Z = \zeta$, and $\mathrm{Var}(Z) = v < \infty$. Dependence of value and cost within a call is unrestricted. Run independent pilot calls and mark each with conditional probability $C_i/N$, using independent randomization. Stop at the second mark, at call $K$. Discard all pilot values and average $M = K - 1$ fresh independent production values. If $\widehat{\zeta}$ is that average and $T$ is the total pilot-plus-production cost, then*

$$\mathbb{E}\widehat{\zeta} = \zeta, \qquad \mathrm{Var}(\widehat{\zeta}) = \frac{vc}{N}, \qquad \mathbb{E}T = 4N - c \le 4N. \tag{39}$$

*The procedure uses neither $c$ nor $v$.*

*Proof.* The pilot marks are independent Bernoulli variables with parameter $p = c/N > 0$, so $K$ has negative-binomial probabilities

$$\mathbb{P}(K = k) = (k-1)p^2(1-p)^{k-2}, \quad k \ge 2, \qquad \mathbb{E}K = 2/p.$$

In particular $K < \infty$ almost surely, and the classical inverse-binomial identity [8] is immediate:

$$\mathbb{E}\frac{1}{K-1} = p^2 \sum_{k=2}^{\infty} (1-p)^{k-2} = p. \tag{40}$$

The event $\{K \ge i\}$ depends only on earlier calls and marks, so is independent of $C_i$. Tonelli's theorem yields

$$\mathbb{E}\sum_{i=1}^{K} C_i = \sum_{i \ge 1} \mathbb{P}(K \ge i)c = c\mathbb{E}K = 2N.$$

Production calls are independent of the pilot. Their mean total cost is $c\mathbb{E}(K-1) = 2N - c$, and conditional on $K$ the production average has mean $\zeta$ and variance $v/(K-1)$. Total expectation and variance give (39). For $p = 1$, $K = 2$ deterministically and the same identities hold. □

## 5.2 Distribution-blind upper bounds

For $B \ge 4$, choose the largest dyadic $N \le B/4$, set $L = \log_2 N$, and apply lemma 9 to $(Z, C) = (Y_L, \mathcal{W}_L)$. By (38),

$$\mathbb{E}T \le 4N \le B, \qquad \mathbb{E}(\widehat{\theta} - \theta)^2 \le \omega_\mu(h)^2 + \frac{85 \cdot 149 A_\mu(h)^2}{N} = 12666\omega_\mu(h)^2. \tag{41}$$

Since $N > B/8$, (7) gives the RMS upper bound, for example with $C_R = \sqrt{8 \cdot 12666}$. For $1 \le B < 4$, output $1/2$ with no observations; then $\omega_\mu(B^{-1/2}) = 1$. This proves (10) without any knowledge of $\mu$ or $q$.

For confidence, choose the least odd integer $k \ge (32/9)\log(1/\delta)$; then $k < 5\log(1/\delta)$ for $0 < \delta \le 1/4$. If $B/k \ge 4$, use $k$ independent groups, each with budget $B/k$, and return their median. A group's error exceeds $\sqrt{8}C_R\omega_\mu(\sqrt{k/B})$ with probability at most $1/8$, by its mean-square bound. Hoeffding's inequality bounds the probability that at least half the groups fail by $e^{-9k/32} \le \delta$. Costs add in expectation, and

$$\omega_\mu(\sqrt{k/B}) \le \sqrt{5}\,\omega_\mu\left(\sqrt{\log(1/\delta)/B}\right).$$

If $B/k < 4$, output $1/2$; then $\sqrt{\log(1/\delta)/B} > 1/\sqrt{20}$, and $\omega_\mu(\sqrt{\log(1/\delta)/B}) > 1/\sqrt{5}$. This handles the remaining case with a universal constant.

**Implementation and scope.** Pilot values are discarded; recycling them into the final average would invalidate the independence calculation. Value and cost may be correlated within any call. The random pilot and production counts are unbounded, so the guarantee concerns expected observations, not a run-by-run work cap. Projection of the final finite-budget estimate onto $[0, 1]$ can only reduce its error, but is not used in the variance identity.

### 5.3 Matching lower bounds

For RMS error use (16) with $h = (4\sqrt{B})^{-1}$. By proposition 6, transcript KL is at most $8/225$, so Pinsker's inequality gives total variation at most $2/15$. Thresholding an estimator at $1/2$ gives a test whose error probabilities sum to at most $8R^2/\Delta_h^2$, if its RMS error under both hypotheses is at most $R$. Every test has error sum at least $1 - 2/15 > 1/2$. Hence $R > \Delta_h/4 = \omega_\mu(h)/16 \geq \omega_\mu(B^{-1/2})/64$. Taking the infimum proves the stated weak inequality.

For confidence write $K_\delta = \mathrm{kl}(1-\delta, \delta)$ and take

$$h = \frac{15}{16\sqrt{2}}\sqrt{K_\delta/B}.$$

Then $225K_\delta/(128h^2) = 4B > B$; by proposition 6 no admissible estimator has radius strictly smaller than $\Delta_h/2 = \omega_\mu(h)/8$. On $0 < \delta \leq 1/4$,

$$K_\delta = (1-2\delta)\log\frac{1-\delta}{\delta} \geq \frac{1}{4}\log(1/\delta). \tag{42}$$

Indeed $1 - 2\delta \geq 1/2$ and $1 - \delta \geq \sqrt{\delta}$. Thus $h \geq 15\sqrt{\log(1/\delta)/B}/(32\sqrt{2})$. Apply (7) to obtain the coefficient $15/(256\sqrt{2})$. This completes the proof of theorem 1.

### 5.4 Proof for a stochastic-dominance envelope

The functions $d \mapsto \min\{1, h/d\}$, $d \mapsto \min\{1, (h/d)^4\}$, and $d \mapsto \min\{h^{-1}, d^{-1}\}$ are decreasing. Their expectations for any magnitude law in (12) are therefore bounded by their expectations under $\mu$. The distribution-blind upper algorithm is unchanged; its risk bound is no larger than the envelope value. The lower bound follows by restricting to $\mathcal{G}(\mu)$, which is a subclass. The same reasoning with $d^{-1}$ and theorem 2 proves corollary 3.

**Remark 10** (Why the statement uses error as a function of budget)**.** For highly irregular $\mu$, inverting $\omega_\mu$ at two different constant multiples of $\varepsilon$ need not preserve multiplicative constants in work. Statements such as $B \asymp \omega_\mu^{-1}(\varepsilon)^{-2}$ need a regularity qualification or explicit constants inside the inverse. The modulus bounds (8)–(9) do not have this defect. Polynomial and logarithmic examples below can be inverted with the stated asymptotic orders.

## 6 Proof of the unbiased characterization

Sufficiency follows from lemma 8, whose analytic constants give $VW \leq 85 \cdot 149 I_\mu^2 = 12665 I_\mu^2$. We prove the converse using only two-point inequalities and the stopped KL chain rule; no differentiation of an estimator's expectation is needed.

**Lemma 11** (Two-point variance inequality)**.** *Let $\mathbb{P}, \mathbb{Q}$ be probability measures on a common space, let $H^2 = 1 - \int \sqrt{d\mathbb{P}\, d\mathbb{Q}}$, and let $Y$ be a real statistic with finite second moment under both. Then $\Delta = \mathbb{E}_{\mathbb{P}} Y - \mathbb{E}_{\mathbb{Q}} Y$ satisfies*

$$\Delta^2(1 - 2H^2) \le 4H^2\big(\mathrm{Var}_{\mathbb{P}}\, Y + \mathrm{Var}_{\mathbb{Q}}\, Y\big), \tag{43}$$

*Moreover, $2H^2 \le \mathrm{KL}(\mathbb{P}, \mathbb{Q})$, and the symmetric bound is*

$$8H^2 \le \mathrm{KL}(\mathbb{P}, \mathbb{Q}) + \mathrm{KL}(\mathbb{Q}, \mathbb{P}). \tag{44}$$

*Proof.* Let $p, q$ be densities with respect to $\nu = \mathbb{P} + \mathbb{Q}$, and let $c = (\mathbb{E}_{\mathbb{P}} Y + \mathbb{E}_{\mathbb{Q}} Y)/2$. Then $\Delta = \int (Y - c)(\sqrt{p} - \sqrt{q})(\sqrt{p} + \sqrt{q})\, d\nu$. We use three facts: $\int (\sqrt{p} - \sqrt{q})^2\, d\nu = 2H^2$; $(\sqrt{p} + \sqrt{q})^2 \le 2(p + q)$; and $\mathbb{E}_{\mathbb{P}}(Y - c)^2 = \mathrm{Var}_{\mathbb{P}}\, Y + \Delta^2/4$, with the same identity under $\mathbb{Q}$. Together with the Cauchy–Schwarz inequality they give

$$\Delta^2 \le \int (\sqrt{p} - \sqrt{q})^2\, d\nu \cdot 2 \int (Y - c)^2 (p + q)\, d\nu = 4H^2\Big(\mathrm{Var}_{\mathbb{P}}\, Y + \mathrm{Var}_{\mathbb{Q}}\, Y + \tfrac{1}{2}\Delta^2\Big).$$

Rearranging gives (43). Finally, by Jensen's inequality, $-\frac{1}{2}\mathrm{KL}(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{\mathbb{P}} \log \sqrt{q/p} \le \log(1 - H^2) \le -H^2$. For the symmetric inequality, use the pointwise bound

$$(x - y) \log(x/y) \ge 4(\sqrt{x} - \sqrt{y})^2, \qquad x, y > 0.$$

Indeed, putting $r = \sqrt{x/y}$, the difference is $2y(r - 1)(r + 1)g(r)$, where $g(r) = \log r - 2(r - 1)/(r + 1)$ has $g(1) = 0$ and $g'(r) = (r - 1)^2/[r(r + 1)^2] \ge 0$. Integrating the pointwise bound gives (44), with zero-density cases interpreted by limits. □

*Converse in theorem 2.* Suppose $Y$ is unbiased for every $q$, with finite uniform variance $V$ and expected work $W$. Fix $h > 0$ and $0 < t \le h/8$, and consider

$$q_{\pm}(d) = \tfrac{1}{2} \pm t\varphi_h(d), \qquad \varphi_h(d) = \min\{h^{-1}, d^{-1}\}.$$

Then $\eta(d) = t\varphi_h(d) \le t/h \le 1/8$ and $d\eta(d) \le t$. Theorem 5 and the stopped-transcript argument give both

$$\max\{\mathrm{KL}(\mathsf{P}_+^{\mathcal{A}}, \mathsf{P}_-^{\mathcal{A}}), \mathrm{KL}(\mathsf{P}_-^{\mathcal{A}}, \mathsf{P}_+^{\mathcal{A}})\} \le \kappa_t := \frac{32t^2 W}{(1 - 4t^2/h^2)^2}. \tag{45}$$

Each direction uses its own expected work, bounded by $W$. Unbiasedness gives mean separation $2tA_\mu(h)$. By (44), $H^2 \le \kappa_t/4$. Applying theorem 11 with both variances at most $V$ yields

$$4t^2 A_\mu(h)^2(1 - \kappa_t/2) \le 2\kappa_t V, \qquad A_\mu(h)^2(1 - \kappa_t/2) \le \frac{16VW}{(1 - 4t^2/h^2)^2}.$$

First let $t \downarrow 0$, with $h$ fixed, giving $A_\mu(h)^2 \le 16VW$. Then $A_\mu(h) \uparrow I_\mu$ as $h \downarrow 0$ by monotone convergence. This proves $I_\mu < \infty$ and $VW \ge I_\mu^2/16$. The envelope class contains $\mathcal{G}(\mu)$, so the same lower bound holds there. □

**Sharpness of the universal coefficient.** For $\mu = \delta_d$, $0 < d \le 1/2$, one fresh observation $X \in \{0, 1\}$ gives

$$Y = \frac{X - 1/2 + d}{2d}, \qquad \mathbb{E}Y = q(d), \qquad \sup_q \mathrm{Var}(Y) = \frac{1}{16d^2}, \qquad W = 1. \tag{46}$$

Since $I_\mu = 1/d$, equality holds. Hence $1/16$ cannot be increased in a bound valid for every magnitude law. This does not assert equality for every nondegenerate law, or that the parameter-free construction achieves the optimal constant for each law.

**Uniformity matters.** The impossibility is over the whole sign class at a fixed $\mu$, or over the stated envelope class. It does not say that each single population with infinite inverse moment forbids every efficient unbiased estimator. For a particular known population, the target might already be known. Likewise, a small structured family of sign functions can be easier than an arbitrary measurable $q$.

### 6.1 Why a universal unbiased output must sometimes be negative

**Proposition 12** (No nonnegative unbiased output except at a deterministic endpoint)**.** *If $\mu((0,1/2)) > 0$, no algorithm stopping almost surely for every $q$ and always returning a nonnegative value is unbiased on $\mathcal{G}(\mu)$. For $\mu = \delta_{1/2}$ the obstruction disappears: one observation is an unbiased $[0,1]$-valued estimator with unit work.*

*Proof.* Under $q \equiv 0$ the target is zero, so nonnegativity and unbiasedness force the output to vanish almost surely. Compare this with $q_1(d) = \mathbf{1}\{d < 1/2\}$, whose target is $\mu((0,1/2)) > 0$. Interior coins give positive likelihood to every finite Bernoulli history under both models; endpoint coins have the same deterministic sign zero. Finite stopped transcript laws are equivalent. Intersecting with each $\{T = n\}$ and taking a countable union gives equivalence of the full stopped transcripts. Seeds and any revealed magnitudes have the same distribution. Thus the output is also zero almost surely under $q_1$, a contradiction. If $\mu = \delta_{1/2}$, the first observation is exactly the sign. □

This elementary support argument is not offered as a separate novelty claim. It explains the signed outputs in the construction. Clipping is allowed for a biased finite-accuracy estimator but invalidates exact unbiasedness.

## 7 Consequences for power and logarithmic margins

Let $F(t) = \mu((0,t])$. Stieltjes integration by parts gives, for $0 < h < a$,

$$\omega_\mu(h) = F(h) + h\int_{(h,a]} \frac{1}{d}\,\mu(\mathrm{d}d) = \frac{h}{a} + h\int_h^a \frac{F(t)}{t^2}\,\mathrm{d}t. \tag{47}$$

The formula is valid with atoms. In particular,

$$I_\mu = \frac{1}{a} + \int_0^a \frac{F(t)}{t^2}\,\mathrm{d}t. \tag{48}$$

This integral, not a power exponent alone, determines exact unbiased feasibility.

### 7.1 The polynomial law as a corollary

For $F(t) = (t/a)^\alpha$ on $(0,a]$ and $u = h/a < 1$,

$$\omega_\mu(h) = \begin{cases} (\alpha u - u^\alpha)/(\alpha - 1), & \alpha \neq 1, \\ u(1 - \log u), & \alpha = 1. \end{cases} \tag{49}$$

Thus the main theorem immediately recovers the three regimes. The usual polynomial-margin class allows a more general constant:

$$\mathcal{G}_{\alpha,C} = \{G \text{ on } [0,1] : \mathbb{P}_G(|P - 1/2| \le t) \le Ct^\alpha, \quad 0 < t \le 1/2\}, \qquad C \ge 2^\alpha. \tag{50}$$

The last condition is necessary for nonemptiness.

**Corollary 13** (Sharp polynomial-margin complexity)**.** *Fix $\alpha > 0$ and $C \geq 2^\alpha$. For all sufficiently small $\varepsilon > 0$ and $0 < \delta \leq 1/4$, the smallest uniform expected work for an $(\varepsilon, \delta)$-valid estimator on $\mathcal{G}_{\alpha,C}$ is*

$$\asymp_{\alpha,C} \log(1/\delta) \begin{cases} \varepsilon^{-2/\alpha}, & 0 < \alpha < 1, \\ \varepsilon^{-2} \log^2(1/\varepsilon), & \alpha = 1, \\ \varepsilon^{-2}, & \alpha > 1. \end{cases} \tag{51}$$

*There is an unbiased estimator with uniformly finite variance and expected work on this entire class if and only if $\alpha > 1$. One common estimator works for every such $\alpha, C$, without knowing either parameter.*

*Proof.* Let $\mu_{\alpha,C}$ have CDF $\min\{Ct^\alpha, 1\}$ on $(0, 1/2]$. It is supported on $(0, C^{-1/\alpha}]$, and $\mathcal{G}_{\alpha,C} = \mathcal{G}_{\preceq \mu_{\alpha,C}}$ exactly. Its modulus is (49) with $a$ replaced by $C^{-1/\alpha}$. Apply theorem 1 and corollary 3 and invert the three orders, with constants depending on $(\alpha, C)$. The inverse moment is finite exactly when $\alpha > 1$, so theorem 2 gives both directions of the unbiased dichotomy. Every polynomial envelope also has a finite uniform logarithmic moment; thus the universal estimator is integrable and unbiased even for $\alpha \leq 1$, when finite uniform variance and work cannot both hold. □

### 7.2 Fixed-gap populations and rare-event scope

**Corollary 14** (Gap classes)**.** *For $0 < \Delta \leq 1/2$, put $\mathcal{G}_\Delta = \{G : \mathbb{P}_G(|P - 1/2| < \Delta) = 0\}$. For sufficiently small $\varepsilon$ and $0 < \delta \leq 1/4$, its minimax uniform expected work for $(\varepsilon, \delta)$ accuracy is*

$$\asymp \frac{\log(1/\delta)}{\varepsilon^2 \Delta^2},$$

*with universal comparison constants and a universal small-accuracy cutoff.*

*Proof.* Here $\mathcal{G}_\Delta = \mathcal{G}_{\preceq \delta_\Delta}$ and $\omega_{\delta_\Delta}(h) = \min\{1, h/\Delta\}$. The envelope confidence law gives the result by direct inversion, taking $\varepsilon$ smaller than its universal lower comparison constant. □

This is the balanced-population specialization of Lee–Valiant [1]. Their finer dependence on the positive fraction $\rho$ is important for rare events; it is outside our worst case over all $q$. Our hard pairs keep every sign prior within $1/8$ of $1/2$, so both targets are near $1/2$. An arbitrary-margin law localized to a small target probability is not established here.

### 7.3 New boundary regimes invisible to a single exponent

For $\beta \geq 0$, define an actual CDF on $(0, a]$ by

$$F_\beta(t) = \frac{t/a}{[1 + \log(a/t)]^\beta}. \tag{52}$$

It is increasing, vanishes at zero, and equals one at $a$. With $L_h = 1 + \log(a/h)$, its modulus is exactly

$$\omega_\beta(h) = \frac{h}{a} \begin{cases} 1 + \dfrac{L_h^{1-\beta} - 1}{1 - \beta}, & \beta \neq 1, \\ 1 + \log L_h, & \beta = 1, \end{cases} \qquad 0 < h < a. \tag{53}$$

This follows from (47) by the substitution $u = 1 + \log(a/t)$.

| Magnitude law near zero | Optimal work at fixed confidence | Efficient unbiased estimator |
|---|---|---|
| $F(t) \asymp t^\alpha$, $\alpha < 1$ | $\varepsilon^{-2/\alpha}$ | No |
| $F(t) \asymp t$ | $\varepsilon^{-2}\log^2(1/\varepsilon)$ | No |
| $F(t) \asymp t^\alpha$, $\alpha > 1$ | $\varepsilon^{-2}$ | Yes |
| $F(t) \asymp t/[\log(1/t)]^\beta$, $\beta < 1$ | $\varepsilon^{-2}[\log(1/\varepsilon)]^{2(1-\beta)}$ | No |
| $F(t) \asymp t/\log(1/t)$ | $\varepsilon^{-2}[\log\log(1/\varepsilon)]^2$ | No |
| $F(t) \asymp t/[\log(1/t)]^\beta$, $\beta > 1$ | $\varepsilon^{-2}$ | Yes |

Table 1: Consequences of the modulus law. "Efficient unbiased" means both variance and expected work bounded uniformly over the corresponding sign or envelope class, not a claim for every individual population. For high confidence, multiply work by $\log(1/\delta)$.

**Corollary 15** (Logarithmically refined critical law)**.** *For the fixed magnitude law* (52)*, or its stochastic-dominance envelope class, the optimal expected work at small accuracy $\varepsilon$ is, up to constants depending on $\beta, a$,*

$$\log(1/\delta) \times \begin{cases} \varepsilon^{-2}[\log(1/\varepsilon)]^{2(1-\beta)}, & 0 \le \beta < 1, \\ \varepsilon^{-2}[\log\log(1/\varepsilon)]^2, & \beta = 1, \\ \varepsilon^{-2}, & \beta > 1. \end{cases} \tag{54}$$

*Uniformly finite variance and work for unbiased estimation are possible exactly when $\beta > 1$, in which case $\mathbb{E}D^{-1} = \beta/[a(\beta-1)]$.*

*Proof.* Equation (53) and theorem 1 and corollary 3 give the error law. Its monotone inverse has the displayed orders: if $\omega(h) \asymp h[\log(1/h)]^\gamma$ with fixed $\gamma \ge 0$, then $h \asymp \varepsilon/[\log(1/\varepsilon)]^\gamma$; replacing the log by log log gives the boundary case. Finally,

$$\mathbb{E}D^{-1} = \frac{1}{a}\left(1 + \int_1^\infty u^{-\beta}\,\mathrm{d}u\right),$$

which is finite exactly when $\beta > 1$. Apply theorem 2. □

For $\beta > 1$, this class permits efficient unbiased estimation even though it satisfies no bound $F(t) \le Ct^\alpha$ with any $\alpha > 1$ and finite $C$. Indeed, $t^{1-\alpha}/[1+\log(a/t)]^\beta \to \infty$. The earlier exponent boundary is therefore a corollary about entire polynomial classes, not the general structural criterion.

**An iterated-log example.** The same calculation extends further. Set $L_1(t) = 1 + \log(a/t)$, $L_2(t) = 1 + \log L_1(t)$, and $F(t) = (t/a)/(L_1(t)L_2(t)^\gamma)$ with $\gamma \ge 0$. Then

$$\omega(h) = \frac{h}{a}\left(1 + \int_1^{L_2(h)} v^{-\gamma}\,\mathrm{d}v\right).$$

At $\gamma = 1$, the fixed-confidence work is $\asymp \varepsilon^{-2}[\log\log\log(1/\varepsilon)]^2$, for sufficiently small $\varepsilon$. Unbiased feasibility again changes at $\gamma > 1$. This is an immediate instance of the general integral criterion, not an additional assumption or an independently designed algorithm.

## 8 Persistence under fixed finite-order density smoothness

The arbitrary-sign model makes no smoothness assumption on $G$. We now show that the critical logarithmic penalty is not solely caused by a jump in the density at the threshold.

For an integer $s \ge 1$, write $\|f\|_{C^s} = \max_{0\le k\le s} \|f^{(k)}\|_\infty$.

**Theorem 16** (Smooth-density robustness)**.** *For each fixed integer $s \geq 1$, there exists a finite $M_s$ such that, for every fixed $M \geq M_s$, the class of $C^\infty$ probability densities supported in $[1/4, 3/4]$ and satisfying $\|f\|_{C^s} \leq M$ has minimax expected complexity*

$$\asymp_{s,M} \varepsilon^{-2} \log^2(1/\varepsilon) \log(1/\delta)$$

*for sufficiently small $\varepsilon$ and $0 < \delta \leq 1/4$. No unbiased estimator has both uniformly finite variance and uniformly finite expected work on this class.*

*Proof.* Fix the local cutoff $a_0 = 1/4$. Choose a fixed even $C^\infty$ density $g$ supported strictly inside $(-a_0, a_0)$, with $g \geq c_g > 0$ on $[-a_0/2, a_0/2]$. The centered coordinate is $x = p - 1/2$. For sufficiently small $h > 0$, set

$$t_h = h^{1/(s+1)}, \qquad f_\pm(1/2 + x) = g(x) \left(1 \pm \frac{2hx}{x^2 + t_h^2}\right). \tag{55}$$

The perturbation is odd, so both densities integrate to one. Its absolute size is at most $h/t_h = h^{s/(s+1)}$, which is less than $1/4$ for small $h$, so the densities are nonnegative. The common magnitude density is $2g(d)$ and the conditional sign perturbation is

$$\eta_h(d) = \frac{hd}{d^2 + t_h^2}, \qquad \eta_h(d) \leq \frac{h}{2t_h} \leq \frac{1}{8}, \qquad d\eta_h(d) \leq h.$$

The posterior lemma therefore gives predictive KL at most $(8192/225)h^2$ per observation, uniformly over histories and selected coins. The target separation satisfies

$$\begin{aligned} \theta_+ - \theta_- &= 4h \int_0^{a_0} g(d) \frac{d}{d^2 + t_h^2} \, \mathrm{d}d \\ &\geq 2c_g h \log \frac{(a_0/2)^2 + t_h^2}{t_h^2} \geq c_s h \log(1/h) \end{aligned}$$

for all sufficiently small $h$, with $c_s > 0$; for a fixed bump it can be chosen proportional to $(s+1)^{-1}$. This is not uniform as $s \to \infty$.

For $R_t(x) = x/(x^2 + t^2) = t^{-1} R_1(x/t)$,

$$\|h R_{t_h}^{(k)}\|_\infty \leq C_k h t_h^{-(k+1)} = C_k h^{(s-k)/(s+1)} \leq C_k, \qquad 0 \leq k \leq s.$$

Leibniz's rule bounds every $\|f_\pm\|_{C^s}$ by a fixed $M_s$ independent of $h$. They are individually $C^\infty$ because $t_h > 0$ and $g$ is a smooth compactly supported function. Uniform bounds on derivatives of all orders at once are not asserted.

Choose $h$ of order $\varepsilon / \log(1/\varepsilon)$ with a sufficiently large fixed factor so that the separation exceeds $2\varepsilon$. The stopped-transcript testing argument proves the lower bound. For the upper bound, $\|f\|_\infty \leq M$ implies $\mathbb{P}(|P - 1/2| \leq t) \leq 2Mt$, so corollary 13 applies. The unbiased impossibility follows by averaging a putative uniform finite-variance, finite-work estimator and contradicting the lower bound at fixed confidence as $\varepsilon \downarrow 0$. □

**Which smoothness is being controlled.** This theorem concerns the density of the latent $P$. It is not a statement about smoothness of a regression function $y \mapsto \mathbb{E}[X \mid Y = y]$ in observable covariates. Smooth cross-state structure can enable information sharing, as in kernel methods [7]. Neither this theorem nor the next one rules out such improvements.

The construction does not settle analogous bounds for fixed analytic or Gevrey regularity classes; those require separate assumptions and arguments.

# 9 An observed-outer-state embedding

A latent coin and an observable outer state are different experiments. The following construction supplies a lower bound in the latter without silently identifying the two.

Let $Y$ be uniform on $[0,1]$. The algorithm may observe independent outer states and may also select any state at which to request a fresh inner observation. Let $X \in \{-1,1\}$ and put $\theta = \mathbb{P}(\mathbb{E}[X \mid Y] > 0)$. There is no regularity assumption connecting the conditional means at different outer states.

**Proposition 17** (Finite-cell observable-state lower bound)**.** *Fix a probability law $\mu$ on $(0,1/2]$, $0 < h \le 1/2$, and $0 < \delta \le 1/8$. There is a finite partition of $[0,1]$ and a family of deterministic, cellwise-constant conditional Bernoulli models such that: (a) their magnitude laws obey the envelope $F_\mu$; (b) their construction has a scale separation $\Delta \ge \omega_\mu(h)/8$; and (c) every estimator valid to radius $\Delta/4$ and failure probability $\delta$ uniformly over this family has worst-case expected work at least*

$$\frac{225}{128h^2}\,\mathrm{kl}(1-2\delta, 2\delta). \tag{56}$$

*This lower bound permits arbitrary selected-state queries and revisiting.*

*Proof.* Round each $d \le h$ up to $h$. Round larger $d$ upward to the next value in the grid $h, 2h, 4h, \ldots$, capped at $a$. This gives a finitely supported law $\mu_h$ on $\{d_1, \ldots, d_J\}$, with weights $w_j$, which is stochastically no smaller than $\mu$. Above $h$ the rounding increases a magnitude by at most a factor two; below $h$ the clipped integrand stays one. Hence

$$\omega_{\mu_h}(h) \ge \tfrac{1}{2}\omega_\mu(h).$$

For each $j$, create $M$ cells of lengths $w_j/M$ and assign unknown fixed signs $s_{jk} \in \{0,1\}$. On cell $(j,k)$ let

$$B \mid Y \sim \mathrm{Ber}\big(1/2 + (2s_{jk}-1)d_j\big), \qquad X = 2B - 1.$$

Every deterministic sign array has magnitude law $\mu_h$, meets the envelope, and has target $\theta(s) = \sum_j (w_j/M) \sum_k s_{jk}$.

For the testing proof only, draw signs independently under two product priors with probabilities $1/2 \pm \eta_h(d_j)$ from (16). Their target means are separated by

$$\Delta = \tfrac{1}{4}\omega_{\mu_h}(h) \ge \tfrac{1}{8}\omega_\mu(h).$$

The sum of squared target weights is $\sum_j w_j^2/M \le 1/M$. Weighted Hoeffding therefore gives

$$\mathbb{P}_\pm(|\theta(s) - \mathbb{E}_\pm\theta(s)| > \Delta/8) \le 2\exp(-M\Delta^2/32).$$

Choose an integer $M \ge 32\Delta^{-2}\log(2/\delta)$. Uniform estimation error at most $\Delta/4$, together with the displayed concentration event, makes thresholding at $1/2$ a test with each error at most $2\delta$.

The signs are independent under either prior, so the local posterior lemma applies separately to every cell, regardless of the adaptively selected query points. Each inner observation has predictive KL at most $128h^2/225$. A uniform expected-work bound for deterministic sign arrays also bounds the prior-averaged expected work. Apply the KL chain rule and binary data processing to obtain (56). □

The partition size depends on the desired scale and confidence. This is a minimax construction in a class allowing arbitrarily fine finite partitions, not an asymptotic lower bound for one fixed finite list of cells. Since the outer distribution itself stays uniform on $[0,1]$, it establishes the relevant obstruction even with observable covariates, provided no cross-state regularity is imposed. The original upper algorithm can ignore the labels and use fresh outer states, so the envelope modulus remains the matching error scale for this unstructured class.

### 9.1 A continuous-magnitude subclass satisfying the nested assumptions

**Proposition 18** (Continuous-magnitude observed-state obstruction)**.** *Fix the local cutoff* $a_0 = 1/4$. *Consider observed outer states* $U \sim \mathrm{Unif}[0,1]$ *and binary inner observations* $X \in \{-1,1\}$ *with*

$$|\mathbb{E}[X \mid U]| = 2D(U), \qquad D(U) \sim \mathrm{Unif}(0, a_0].$$

*No cross-state regularity is assumed. There is a class of deterministic conditional models of this form whose minimax expected observation cost for error* $\varepsilon$ *with fixed failure probability* $1/16$ *is*

$$\asymp \varepsilon^{-2} \log^2(1/\varepsilon).$$

*The lower bound allows arbitrary selected-state queries and revisiting. All models satisfy the bounded-density and normalized conditional-moment assumptions 2.1 and 2.4 of Giles–Haji-Ali [2], with constants independent of the error scale.*

*Proof.* Fix $0 < h < a_0$ and partition $(0, a_0]$ into the first band $(0, h]$ and consecutive dyadic bands with right endpoints $2h, 4h, \ldots$, clipping the final endpoint at $a_0$. Let band $j$ have endpoints $\ell_j, b_j$ and mass $w_j = (b_j - \ell_j)/a_0$. Partition $[0,1]$ into $M$ equal-weight cells per band, each of length $w_j/M$. Within every cell in band $j$, let $D(U)$ increase linearly across $(\ell_j, b_j]$. Define its values at the finitely many endpoints to be positive and in the assigned band. Consequently the magnitude law is exactly uniform on $(0, a_0]$, not a discretization. The partition and the map $D$ may be disclosed to the algorithm.

Assign each cell an unknown deterministic sign $s_{jk} \in \{0,1\}$. On that cell let $X = 2B - 1$ for $B \sim \mathrm{Ber}(1/2 + (2s_{jk} - 1)D(U))$. The target is

$$\theta(s) = \sum_j \frac{w_j}{M} \sum_{k=1}^{M} s_{jk}.$$

For the proof only, choose independent cell signs under two priors with probabilities $1/2 \pm \eta_j$, where $\eta_j = \frac{1}{8}\min\{1, h/b_j\}$. Their target means have separation

$$\Delta_h = 2\sum_j w_j \eta_j \geq \frac{1}{8}\omega_\mu(h), \qquad \mu = \mathrm{Unif}(0, a_0].$$

Indeed, within the first band the clipped integrand is one; within every other band $b_j \leq 2D(U)$. Every queried magnitude in band $j$ satisfies $D(U)\eta_j \leq h/8$.

Repeated queries in a cell may use different magnitudes. Nevertheless, its sign has a single common likelihood ratio under the two priors, obtained by multiplying its queried sign-conditional likelihoods. The posterior identity and the predictive bound therefore still apply. Each observation contributes at most $128h^2/225$ KL, even under arbitrary query selection.

Weighted Hoeffding gives

$$\mathbb{P}_\pm\big(|\theta(s) - \mathbb{E}_\pm\theta(s)| > \Delta_h/8\big) \leq 2e^{-M\Delta_h^2/32}.$$

For $M \geq 32\Delta_h^{-2}\log(2/\delta)$, an estimator with uniform radius $\Delta_h/4$ and failure probability $\delta \leq 1/8$ gives a test of these priors with each error at most $2\delta$. The stopped-transcript argument gives expected work at least

$$\frac{225}{128h^2}\,\mathrm{kl}(1 - 2\delta, 2\delta).$$

For the uniform magnitude law, $\omega_\mu(h) = (h/a_0)[1 + \log(a_0/h)]$. Set $h$ of order $\varepsilon/\log(1/\varepsilon)$ with a sufficiently large fixed constant to obtain the lower rate. The upper rate follows by ignoring state labels and using the preceding threshold estimator. The class permits arbitrarily many cells; it is not a fixed finite-partition asymptotic claim.

Finally, $\mathrm{Var}(X \mid U) = 1 - 4D(U)^2 \geq 3/4$. Their normalized magnitude is

$$Z = \frac{|\mathbb{E}[X \mid U]|}{\sqrt{\mathrm{Var}(X \mid U)}} = \frac{2D}{\sqrt{1 - 4D^2}}.$$

It has density $[2a_0(1 + z^2)^{3/2}]^{-1}$ on its support, bounded by $1/(2a_0)$. Moreover, $|X - \mathbb{E}[X \mid U]| \leq 3/2$, so for every finite $q > 2$ its standardized conditional $q$th absolute moment is bounded by $3^{q/2}$. These verify the two stated assumptions uniformly. A uniform RMS guarantee $\varepsilon$ gives error at most $4\varepsilon$ with failure probability at most $1/16$, by Markov's inequality. Applying the fixed-confidence lower bound at $4\varepsilon$ yields the same RMS rate. The RMS upper bound follows from theorem 1. This also proves corollary 4. □

**Interpretation.** This identifies a hard subclass satisfying the explicit density and moment assumptions behind the classical adaptive nested-simulation upper rate. It does not rule out improvements from smooth cross-state regression structure, a known parametric model, or restrictions to rare-event populations. A restriction on the density of the latent magnitude is different from a restriction on the function of observed covariates.

## 10 Computational verification and limitations

The reference implementation includes the coupled dyadic classifiers, geometric-level estimator, and two-success distribution-blind calibration. The deterministic classifier check propagates shared-prefix success-count probabilities and both stopping decisions. It uses floating-point arithmetic, not Monte Carlo, and checks the coupled increment rather than only marginal classification errors. Symbolic and high-precision checks cover the posterior identity, both predictive-KL directions, the symmetric KL–Hellinger inequality, and the inverse-binomial identities.

| Diagnostic | Count | Result / maximum observed |
|---|---|---|
| Symbolic algebraic identities | 7 | Exact simplifications |
| Predictive KL, both directions | 630 | Bound ratio $< 1$ |
| Hard-pair directional KL | 280 | $\mathrm{KL}/h^2 \leq 0.503802$ |
| Symmetric KL–Hellinger checks | 315 | $8H^2 \leq \mathrm{KL} + \mathrm{KL}^{\mathrm{rev}}$ |
| Coupled classifier dynamic programs | 264 | 11 magnitudes, both signs |
| Classifier error / bound | | 0.748999 |
| Second moment / truncated inverse gap | | 3.414214 (proved 85) |
| Expected work / truncated inverse gap | | 2.931912 (proved 149) |
| Maximum probability-mass residual | | $4.22 \times 10^{-15}$ |
| Maximum telescoping residual | | $2.11 \times 10^{-15}$ |
| Modulus integral / closed-form checks | 96 | Residual $\leq 2.23 \times 10^{-16}$ |

Table 2: Recomputed finite diagnostics for v4. Predictive and Hellinger comparisons use 120-decimal arithmetic. Classifier dynamic programs use floating-point probability recursion with caps through $2^{11}$ and include $D = 1/2$; their maxima are not replacements for analytic constants.

The calibration implementation is separately exercised on synthetic value–cost distributions, including dependent values and costs, zero-cost calls, and the deterministic-mark endpoint.

These Monte Carlo diagnostics are not universal bounds or accuracy benchmarks. The independent production sample is retained in every implementation. Reproduction commands and full outputs are in the source package; none of the reported checks is a proof of a theorem over an infinite domain.

**Limitations.** The observation model requires independent fresh coins and conditionally independent observations of each fixed coin. Dependence across new tasks, drifting probabilities, unknown noise mechanisms, and heterogeneous verifiers are not covered. The theorems optimize expected observation work; they do not provide deterministic latency bounds or a data-computable confidence radius over all unknown margin laws. The unbiased estimator and calibration wrapper terminate almost surely but can have rare, large work values. No practical speedup on real verifiers or financial portfolios is claimed. Analytic upper constants remain conservative and have not been replaced by favorable finite-grid maxima.

The minimax lower bounds use balanced sign populations and no cross-state regularity. A target-localized rare-event characterization, optimal cost-tail bounds, and fixed analytic/Gevrey classes are separate questions. Correctness and publication priority require independent specialist review.

## 11 Conclusion

The clipped inverse-margin modulus $\mathbb{E}\min\{1, h/D\}$ characterizes minimax adaptive threshold error for every nonzero magnitude law under an expected observation budget. A single distribution-blind rule attains this rate for all such laws, using classical inverse-binomial calibration to avoid a known-cost allocation. The same conditional estimates identify $\mathbb{E}[D^{-1}] < \infty$ as the exact boundary for the conjunction of uniformly finite variance and expected work in unbiased estimation. The symmetric-KL converse gives the sharp universal lower coefficient $1/16$, with equality for fixed-gap laws.

An observed-state hard subclass verifies the density and conditional-moment assumptions used in adaptive nested simulation and exhibits the unavoidable critical squared logarithm. The result does not preclude improvements from additional structure or different information. Practical implementation requires model validation and attention to cost tails, while scholarly assessment requires a dedicated proof and priority review.

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